

The empirical recurrence for OEIS A237845: recovering a conjecture that is not written down

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Abstract

OEIS A237845 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 171$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237845 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum minus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

1697, 31661, 581994, 11700943, 234481287, 4792713430, 97664853059, 2001409956781, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:30:27, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 171$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 171$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 342$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	29	c_{19}	27903095438170592	c_{37}	−872721691626771756306528	c_{55}	40323
c_2	212	c_{20}	612375987544108578	c_{38}	−314621596281858441727192	c_{56}	−5674
c_3	−11594	c_{21}	−675647143703665928	c_{39}	1868130830625283825144686	c_{57}	−11422
c_4	16048	c_{22}	−6647034900341286560	c_{40}	369683330602431932092578	c_{58}	13092
c_5	1851803	c_{23}	9872761858009910305	c_{41}	−3260154604117915151179776	c_{59}	23968
c_6	−8142699	c_{24}	57019345037769166467	c_{42}	−244740435375417184187156	c_{60}	−1987
c_7	−161957175	c_{25}	−103561560629448744210	c_{43}	4635396789184137041538152	c_{61}	−3567
c_8	1018196013	c_{26}	−389046410915450979941	c_{44}	−75988235571601561857080	c_{62}	1799
c_9	8759605120	c_{27}	826802405921981697834	c_{45}	−5358219727367962615339600	c_{63}	3529
c_{10}	−70791362382	c_{28}	2119446784165876553094	c_{46}	441055859959968374913904	c_{64}	−82
c_{11}	−308024111873	c_{29}	−5167503124808330098395	c_{47}	5016597735990622931562176	c_{65}	−209
c_{12}	3226028798774	c_{30}	−9230550030878199073073	c_{48}	−635886941330459459715136	c_{66}	19
c_{13}	6991536735908	c_{31}	25695519904846129976904	c_{49}	−3782944029707679206804928	c_{67}	629
c_{14}	−103773108838562	c_{32}	32082555344956204608426	c_{50}	577437197600364143509632	c_{68}	−9
c_{15}	−88876774693040	c_{33}	−102696948207908747745380	c_{51}	2279773917855028474530560	c_{69}	−6
c_{16}	2459509894134199	c_{34}	−88470689861335223816157	c_{52}	−369836611493021336015872	c_{70}	2
c_{17}	−29718656767235	c_{35}	332106805798664891973145	c_{53}	−1086273458889201801969664		
c_{18}	−44162259405169850	c_{36}	191092255163057612377480	c_{54}	171111626468250107523072		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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